

Question				Marking details	Marks Available																																																					
					AO1	AO2	AO3	Total	Maths	Prac																																																
4	(a)			Two or more polypeptide chains {bonded/ joined} together (to form a functional protein)	1			1																																																		
	(b)	(i)		438 (Accept 441/444)		1		1	1																																																	
		(ii)		A triplet code can produce 64 different combinations, which is enough to code for the 20 different amino acids/ 2 bases would only give 16 combinations which is not enough for 20 amino acids (1) (As there are more than 20 codes) some amino acids are coded for by more than one mRNA codon. (1)	2			2																																																		
		(iii)		<table border="1"><thead><tr><th></th><th colspan="7">Codon position number</th></tr><tr><th></th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th colspan="2">8</th></tr></thead><tbody><tr><td>DNA coding strand</td><td>GAC</td><td>TGA</td><td>GGA</td><td>CTC</td><td>CTC</td><td>TTC</td><td>TTT</td></tr><tr><td>mRNA codons</td><td>CUG</td><td>ACU</td><td>CCU</td><td>GAG</td><td>GAG</td><td>AAG</td><td>AAA</td></tr><tr><td>tRNA anticodons</td><td>GAC</td><td>UGA</td><td>GGA</td><td>CUC</td><td>CUC</td><td>UUC</td><td>UUU</td></tr><tr><td>Amino acid sequence</td><td>leucine</td><td>threonine</td><td>proline</td><td>Glutamic acid</td><td>Glutamic acid</td><td>lysine</td><td>lysine</td></tr></tbody></table> Two marks for 6 correct cells in columns 3,4 and 5. one mark for 4/5 correct cells (2) One mark for correct amino acid sequence (1) One mark for column 8 correct (1)		Codon position number								3	4	5	6	7	8		DNA coding strand	GAC	TGA	GGA	CTC	CTC	TTC	TTT	mRNA codons	CUG	ACU	CCU	GAG	GAG	AAG	AAA	tRNA anticodons	GAC	UGA	GGA	CUC	CUC	UUC	UUU	Amino acid sequence	leucine	threonine	proline	Glutamic acid	Glutamic acid	lysine	lysine		4		4		
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	(c)			Any five (x1) from: A. Translation (1) B. tRNA brings an amino acid to the ribosome (1) C. Each tRNA has a {specific/ particular} amino acid (1) D. Complementary base pairing occurs between the tRNA anticodon and the mRNA codon/ description of codon anticodon interaction (1) E. A second amino acid is brought to the ribosome / two tRNA molecules can attach to the mRNA (at the ribosome)/ reference to two binding sites. (1) F. A condensation reaction occurs/ a peptide bond forms between the amino acids. (1) G. The ribosome moves along the mRNA {by one codon/until a stop codon is reached.} (1)	5			5		
	(d)			{Thymine/ T} is replaced with {adenine/ A} / CAC. (1)		1		1		
	(e)	(i)		Both R groups circled	1			1		
		(ii)		Phenylalanine, leucine and valine are all non- polar and therefore are attracted to each other (1) Glutamic acid is polar and will therefore {be repelled from/ not attracted to} phenylalanine and leucine (1) any correct reference to hydrophobic/ hydrophilic (1)			3	3		
				Question 4 total	9	6	3	18	1	0